Week 4 - Part E: Justification of the e-commerce review figure

CDB601220 Data Visualization. All data are synthetic course data (Superior University Lahore), not findings about a real business.

1. Per panel: the question, the chart type, the Week 2 channel rule

Panel 1 - 'Where is traffic really heading?' A line over time: time on the x axis and sessions on a common aligned scale, i.e. **position**, the most accurate channel (position > length > angle > area). Raw daily totals stay in **grey for context**; the 7-day rolling mean is the single coloured **pop-out**, so the eye finds the trajectory, not the weekend rhythm.

Panel 2 - 'When in the week do people visit?' A day-of-week x hour heatmap: two categorical axes give position, magnitude goes to colour lightness. Colour is a weaker channel than position, so it is used for the pattern (where is the dark band?), not for exact values. One-hue **sequential** colormap because there is no midpoint (Week 3).

Panel 3 - 'Where is the mass?' A treemap: containment shows category -> sub-category and **area** (the 4th-best channel, read to about +/-20%) shows size, so every block carries its value in text. One hue per parent category and only 4 parents respect the 4 +/-1 chunk limit.

Panel 4 - 'By how much?' The same eight numbers as sorted horizontal bars: **position on a common scale**, values printed, rank readable top to bottom. Panels 3 and 4 answer different questions about the same data: where the mass is, and by how much.

2. Every time decision, named and defended

Parse. The file mixes 9,019 ISO timestamps and 3,651 slash timestamps. Each is parsed with its own explicit format. The slash format is **day-first**: its first field reaches 31 while its second never exceeds 12, so month-first is impossible. `format='mixed'` parses every row without an error but moves 1,382 rows to a wrong date and stretches the span from 528 to 704 days. I checked min and max, not just the row count: 2024-04-01 to 2025-09-11.

Resolution and aggregation. Hourly rows are resampled to **daily with sum**, because sessions are a **flow** (visits that began during the hour); a flow is never resampled with last(). The axis says 'Sessions per day (sum of hourly)'. Gaps were counted on the hourly index *before* resampling, and `sum(min_count=24)` makes an incomplete day NaN rather than a silent short total or a zero.

Window. 7 days, centred, requiring at least 5 of 7 days present. It cancels the weekly cycle exactly (Part B.1: removes 31.7% of raw variation, loses 3 days at each end; 28 days would cost 14, 91 days 45). **Centred, not trailing**, because this is a retrospective Monday review: a centred mean puts turns on the right date, while a trailing mean would report them about half a window (3 days) late. A live dashboard would need the trailing version and a caption saying so.

Ethics clause. The window was fixed at 7 days before plotting, because the cycle being removed is weekly (Part B.1). The series is shown from its first recorded day (1 April 2024); no start date was selected. A 28-day version is in the appendix and does not change the direction (the 7- and 28-day lines correlate 0.96). Part B.5 shows why this matters: on the same series 34 of 60 start-and-window combinations rise and 26 fall. Symmetry test: I would have kept the same 7-day window if it had weakened my conclusion, because the weekly cycle, not the picture, sets it.

3. Every row removed (rule, count, what moved)

Item	Rule	Count	What moved
Duplicate hours	keep first row per timestamp; every pair verified value-identical	286 rows (12,670 -> 12,384)	total sessions -2.36%
Sentinel sessions = -1	set to NaN; daily totals refilled with same weekday-hour median, flagged	0 rows removed; 287 hours refilled	adds 2.33% to daily total; excluded from heatmap
Days never recorded	left as gaps (NaN), never zero-filled	0 rows; 13 days / 312 hours absent	line breaks at 13 places
Partial month (2025-09)	kept, shaded and labelled partial; never compared as a total	0 rows; 11 of 30 days	excluded from monthly-median statements
Duplicate SKU (SKU-0127)	keep the copy whose sku_name prefix matches its sub-category	1 row (345 -> 344)	Wearables share 10.13% -> 9.35%; Audio 14.86% -> 14.99%
Negative net revenue	never drawn as area; netted inside sub-category totals and reported	0 rows; 7 SKUs, -1.07m in total	all 8 sub-category totals stay positive

4. Every level rolled up

No 'Other' bucket was needed: 0 categories folded, value 0m, because the smallest sub-category (Drones) is still 2.4% of the total. The depth was cut at **sub-category**: 344 SKUs were summed into 8 sub-categories (total net revenue 1,711.4m), because 344 blocks would be unreadable. Tree checks passed: every sub-category has exactly one parent and the children sum to their parents. Self-check against shares_eight.csv: every sub-category share is within 0.3 percentage points. In Panel 2, 168 weekday-hour cells are shown unrolled.